until sterile ‘soldiers’ were discovered in a Japanese species of aphids by Shigeyuki Aoki, in 1977, just too late to appear in the first edition of this book. Aoki has since found the phenomenon in a number of different species, and has good evidence that it has evolved at least four times independently in different groups of aphids.

Briefly, Aoki’s story is this. Aphid ‘soldiers’ are an anatomically distinct caste, just as distinct as the castes of traditional social insects like ants. They are larvae that do not mature to full adulthood, and they are therefore sterile. They neither look nor behave like their non-soldier larval contemporaries, to whom they are, however, *genetically* identical. Soldiers are typically larger than non-soldiers; they have extra-big front legs which make them look almost scorpion-like; and they have sharp horns pointing forward from the head. They use these weapons to fight and kill would-be predators. They often die in the process, but even if they don’t it is still correct to think of them as genetically ‘altruistic’ because they are sterile.

In terms of selfish genes, what is going on here? Aoki does not mention precisely what determines which individuals become sterile soldiers and which become normal reproductive adults, but we can safely say that it must be an environmental, not a genetic difference—obviously, since the sterile soldiers and the normal aphids on any one plant are genetically identical. However, there must be genes for the capacity to be environmentally switched into either of the two developmental pathways. Why has natural selection favoured these genes, even though some of them end up in the bodies of sterile soldiers and are therefore not passed on? Because, thanks to the soldiers, copies of those very same genes have been saved in the bodies of the reproductive non-soldiers. The rationale is just the same as for all social insects (see Chapter 10), except that in other social insects, such as ants or termites, the genes in the sterile ‘altruists’ have only a *statistical* chance of helping copies of themselves in non-sterile reproductives. Aphid altruistic genes enjoy certainty rather than statistical likelihood since aphid soldiers are clone-mates of the reproductive sisters whom they benefit. In some respects Aoki’s aphids provide the neatest real-life illustration of the power of Hamilton’s ideas.

Should aphids, then, be admitted to the exclusive club of truly social insects, traditionally the bastion of ants, bees, wasps, and termites? Entomological conservatives could blackball them on various grounds. They lack a long-lived old queen, for instance. Moreover, being a true clone, the aphids are no more ‘social’ than the cells of your body. There is a single animal feeding on the plant. It just happens to have its body divided up into physically separate aphids, some of which play a specialized defensive role just like white blood corpuscles in the human body. ‘True’ social insects, the argument goes, cooperate in spite of not being part of the same organism, whereas Aoki’s aphids cooperate because they do belong to the same ‘organism’. I cannot get worked up about this semantic issue. It seems to me that, so long as you understand what is going on among ants, aphids, and human cells, you should be at liberty to call them social or not, as you please. As for my own preference, I have reasons for calling Aoki’s aphids social organisms, rather than parts of a single organism. There are crucial properties of a single organism which a single aphid possesses, but which a clone of aphids does not possess. The argument is spelled out in *The Extended Phenotype*, in the chapter called ‘Rediscovering the Organism’, and also in the new chapter of the present book called ‘The long reach of the gene’.

p. 122 Kin selection is emphatically not a special case of group selection.

The confusion over the difference between group selection and kin selection has not disappeared. It may even have got worse. My remarks stand with redoubled emphasis except that, by a thoughtless choice of words, I introduced a quite separate fallacy of my own at the top of page 102 of the first edition of this book. I said in the original (it is one of the few things I have altered in the text of this edition): ‘We simply expect that second cousins should tend to receive as much altruism as offspring or siblings.’ As S. Altmann has pointed out, this is obviously wrong. It is wrong for a reason that has nothing to do with the point I was trying to argue at the time. If an